ANSWERS

1.

(a)	**(i)**	packet/discrete quantity/quantum (of energy) of e.m. radiation	B1
	(ii)	either $E = (6.63 \times 10^{-34} \times 3 \times 10^{8})/(350 \times 10^{-9})$ or $E = (6.63 \times 10^{-34} \times 8.57 \times 10^{14})$ $E = 5.68 \times 10^{-19}$ J	 M1 A0
	(iii)	0.5	B1
(b)	**(i)**	energy of photon to cause emission of electron <u>from surface</u> *either* with zero k.e *or* photon energy is minimum	M1 A1
	(ii)	correct conversion eV $\rightarrow$ J or J $\rightarrow$ eV seen once photon energy must be greater than work function 350 nm wavelength and potassium metal	B1 C1 A1

2.

(a) e.g. 'instantaneous' emission (of electrons)
threshold frequency below which no emission
(max) <u>electron</u> energy dependent on frequency
(max) <u>electron</u> energy not dependent on intensity
rate of emission (of electrons) depends on intensity
(*any three sensible suggestions, 1 each*) B3

(b)	**(i)**	'packet' / quantum of energy of electromagnetic energy / radiation	M1 A1
	(ii)	discrete wavelengths mean photons have particular energies energy of photon determined by energy change of (orbital) electron so discrete energy levels	M1 M1 A0
(c)	**(i)**	three energy changes shown correctly arrows 'pointing' in correct direction wavelengths correctly identified	B1 B1 B1
	(ii)	chooses λ = 486 nm $\Delta E = hc / \lambda$ $= (6.63 \times 10^{-34} \times 3.0 \times 10^{8}) / (4.86 \times 10^{-9})$ $= 4.09 \times 10^{-19}$ J (*allow 2 s.f.*)	C1 C1 A1

3.

(a) wave theory predicts any frequency would give rise to emission of electron M1
if exposure time is sufficiently long A1
photon has (specific value of) energy dependent on frequency M1
emission if energy greater than threshold / work function / energy to remove electron from surface A1

(b) photon is packet/quantum of energy M1
of electromagnetic radiation A1
(photon) energy = h × frequency B1

every <u>particle</u> has an (associated) wavelength B1
wavelength = h / p M1
where p is the momentum (of the particle) A1

4.

(a) each line corresponds to a (specific) photon energy B1
photon emitted when electron changes its energy level B1
discrete energy changes so discrete levels B1

(b) **(i)** $E = hc / \lambda$ …(*allow ratio ideas*) C1
$= (6.63 \times 10^{-34} \times 3.0 \times 10^{8}) / (486 \times 10^{-9})$
$= 4.09 \times 10^{-19}$ J A1

(ii) four transitions to/from -5.45×10^{-19} J level B1
all transitions shown from higher to lower energy (level) B1

5.

(a) **(i)** e.g. electron / particle diffraction B1

(ii) e.g. photoelectric effect B1

(b) **(i)** 6 A1

(ii) change in energy = 4.57×10^{-19} J
$\lambda = hc / E$ C1
$= (6.63 \times 10^{-34} \times 3.0 \times 10^{8}) / (4.57 \times 10^{-19})$
$= 4.4 \times 10^{-7}$ m A1

6.

(a)	minimum frequency for electron to be emitted (from surface)	M1
	of electromagnetic radiation / light / photons	A1
(b)	$E = hc / \lambda$ *or* $E = hf$ <u>and</u> $c = f\lambda$	C1
	either threshold wavelength $= (6.63 \times 10^{-34} \times 3.0 \times 10^{8}) / (5.8 \times 10^{-19})$	
	$= 340$ nm	
	or energy of 340 nm photon $= 4.4 \times 10^{-19}$ J	
	or threshold frequency $= 8.7 \times 10^{14}$ Hz	
	or 450 nm $\rightarrow 6.7 \times 10^{14}$ Hz	A1
	appropriate comment comparing wavelengths / energies / frequencies	B1
	so no effect on photo-electric current	B1

7.

(a)	for a wave, electron can 'collect' energy continuously	B1
	for a wave, electron will always be emitted / electron will be emitted at all frequencies…..	M1
	after a sufficiently long delay	A1
(b) (i)	*either* wavelength is longer than threshold wavelength	
	or frequency is below the threshold frequency	
	or photon energy is less than work function	B1
(ii)	$hc / \lambda = \phi + E_{MAX}$	C1
	$(6.63 \times 10^{-34} \times 3.0 \times 10^{8}) / (240 \times 10^{-9}) = \phi + 4.44 \times 10^{-19}$	C1
	$\phi = 3.8 \times 10^{-19}$ J (*allow* 3.9×10^{-19} J)	A1
(c) (i)	photon energy larger	M1
	so (maximum) kinetic energy is larger	A1
(ii)	fewer photons (per unit time)	M1
	so (maximum) current is smaller	A1

8.

(a)	wavelength of wave associated with a particle	M1
	that is moving	A1
(b) (i)	energy of electron $= 850 \times 1.6 \times 10^{-19}$	M1
	$= 1.36 \times 10^{-16}$ J	
	energy $= p^2 / 2m$ *or* $p = mv$ <u>and</u> $E_K = \frac{1}{2}mv^2$	
	momentum $= \sqrt{(1.36 \times 10^{-16} \times 2 \times 9.11 \times 10^{-31})}$	M1
	$= 1.6 \times 10^{-23}$ N s	A0

(ii) $\lambda = h / p$ C1

wavelength $= (6.63 \times 10^{-34}) / (1.6 \times 10^{-23})$

$= 4.1 \times 10^{-11}$ m A1

(c) diagram or description showing:

electron beam in a vacuum B1

incident on thin metal target / carbon film B1

fluorescent screen B1

pattern of concentric rings observed M1

pattern similar to diffraction pattern observed with visible light A1

9.

(a) each line represents photon of specific energy M1

photon emitted as a result of energy change of electron M1

specific energy changes so discrete levels A1

(b) **(i)** arrow from –0.85 eV level to –1.5 eV level B1

(ii) $\Delta E = hc / \lambda$ C1

$= (1.5 - 0.85) \times 1.6 \times 10^{-19}$ C1

$= 1.04 \times 10^{-19}$ J

$\lambda = (6.63 \times 10^{-34} \times 3.0 \times 10^{8})/(1.04 \times 10^{-19})$

$= 1.9 \times 10^{-6}$ m A1

(c) spectrum appears as continuous spectrum crossed by dark lines B1

two dark lines B1

electrons in gas absorb photons with energies equal to the excitation energies M1

light photons re-emitted in all directions A1

10.

(a) **(i)** packet/quantum of energy M1

of electromagnetic radiation A1

(ii) minimum energy to cause emission of an electron (from surface) B1

(b) **(i)** $hc/\lambda = \Phi + E_{max}$ M1

c and *h* explained A1

(ii) **1.** *either* when $1/\lambda = 0$, $\Phi = -E_{max}$
or evidence of use of x-axis intercept from graph
or chooses point close to the line and substitutes values of $1/\lambda$ and E_{max} into $hc/\lambda = \Phi + E_{max}$ C1
$\Phi = 4.0 \times 10^{-19}$ J (allow $\pm 0.2 \times 10^{-19}$ J) A1

2. *either* gradient of graph is $1/hc$ C1
gradient $= 4.80 \times 10^{24} \rightarrow 5.06 \times 10^{24}$ M1
$h = 1/(\text{gradient} \times 3.0 \times 10^{8})$
$= 6.6 \times 10^{-34}$ J s $\rightarrow$ 6.9×10^{-34} J s A1
or chooses point close to the line and substitutes values of $1/\lambda$ and E_{max} into $hc/\lambda = \Phi + E_{max}$ (C1)
values of $1/\lambda$ and E_{max} are correct within half a square (M1)
$h = 6.6 \times 10^{-34}$ J s $\rightarrow$ 6.9×10^{-34} J s (A1)
(Allow full credit for the correct use of any appropriate method)
(Do not allow 'circular' calculations in ***part 2*** *that lead to the same value of Planck constant that was substituted in* ***part 1****)*

11.

(a) minimum energy to remove an electron from the metal/surface B1

(b) gradient $= 4.17 \times 10^{-15}$ *(allow 4.1 → 4.3)* C1
$h = 4.15 \times 10^{-15} \times 1.6 \times 10^{-19}$ *or* $h = 4.1$ to 4.3×10^{-15} eV s A1
$= 6.6 \times 10^{-34}$ J s A0

(c) graph: straight line parallel to given line
with intercept at any higher frequency B1
intercept at between 6.9×10^{14} Hz and 7.1×10^{14} Hz B1

12.

(a) discrete quantity / packet / quantum of energy of electromagnetic radiation B1
energy of photon = Planck constant × frequency B1

(b) threshold frequency (1)
rate of emission is proportional to intensity (1)
max. kinetic energy of electron dependent on frequency (1)
max. kinetic energy independent of intensity (1)
(*any three, 1 each, max 3*) B3

(c)	*either* $E = hc/\lambda$	*or* $hc/\lambda = eV$	C1
	λ = 450 nm to give	work function of *3.5 eV*	
	energy = 4.4×10^{-19} or 2.8 eV	to give λ = 355 nm	M1
	2.8 eV < 3.5 eV so no emission	355 nm < 450 nm so no	A1

or work function = 3.5 eV
threshold frequency = 8.45×10^{14} Hz — C1
450 nm = 6.67×10^{14} Hz — M1
6.67×10^{14} Hz < 8.45×10^{14} Hz — A1

13.

(a) wavelength associated with a particle — M1
that is moving — A1

(b) (i) kinetic energy = $1.6 \times 10^{-19} \times 4700$ — C1
$= 7.52 \times 10^{-16}$ J
either energy = $p^2/2m$ *or* $E_K = \frac{1}{2}mv^2$ *and* $p = mv$ — C1
$p = \sqrt{(7.52 \times 10^{-16} \times 2 \times 9.1 \times 10^{-31})}$ — C1
$= 3.7 \times 10^{-23}$ N s
$\lambda = h/p$ — C1
$= (6.63 \times 10^{-34}) / (3.7 \times 10^{-23})$
$= 1.8 \times 10^{-11}$ m — A1

(ii) wavelength is about separation of atoms — B1
can be used in (electron) diffraction — B1

14.

(a) (i) lowest frequency of e.m. radiation — M1
giving rise to emission of electrons (from the surface) — A1

(ii) $E = hf$ — C1
threshold frequency $= (9.0 \times 10^{-19}) / (6.63 \times 10^{-34})$
$= 1.4 \times 10^{15}$ Hz — A1

(b) *either* 300 nm ≡ 10×10^{15} Hz (and 600 nm ≡ 5.0×10^{14} Hz)
or 300 nm ≡ 6.6×10^{-19} J (and 600 nm ≡ 3.3×10^{-19} J)
or zinc λ_0 = 340 nm, platinum λ_0 = 220 nm (and sodium λ_0 = 520 nm) — M1
emission from sodium <u>and</u> zinc — A1

(c) each photon has larger energy — M1
fewer photons per unit time — M1
fewer electrons emitted per unit time — A1

15.

(a) each wavelength is associated with a discrete <u>change</u> in energy — M1
discrete energy <u>change</u> / difference implies discrete levels — A1

(b) **(i)** **1.** arrow from –0.54 eV to –0.85 eV, labelled L — B1

2. arrow from –0.54 eV to –3.4 eV , labelled S — B1
(two correct arrows, but only one label – allow 2 marks)
(two correct arrows, but no labels – allow 1 mark)

(ii) $E = hc / \lambda$ — C1
$(3.4 - 0.54) \times 1.6 \times 10^{-19} = (6.63 \times 10^{-34} \times 3.0 \times 10^{8}) / \lambda$ — C1
$\lambda = 4.35 \times 10^{-7}$ m — A1

(c) $-1.50 \rightarrow -3.4 = 1.9$ eV
$-0.85 \rightarrow -3.4 = 2.55$ eV (allow 2.6 eV)
$-0.54 \rightarrow -3.4 = 2.86$ eV (allow 2.9 eV)
3 correct, 2 marks with –1 mark for each additional energy
2 correct, 1 mark but no marks if any additional energy differences — B2

16.

(a) *either* if light passes through suitable film / cork dust etc. — M1
diffraction occurs and similar pattern observed — A1
or concentric circles are evidence of diffraction — (M1)
diffraction is a wave property — (A1)

(b) (speed increases so) momentum increases — M1
$\lambda = h/p$ so λ decreases — M1
hence radii decrease — A1
(*special case: wavelength decreases so radii decreases – scores 1/3*)
or
(speed increases so) energy increases — (B1)
$\lambda = h / \sqrt{(2Em)}$ so λ decreases — (M1)
hence radii decrease — (A1)

(c) electron and proton have same (kinetic) energy — C1
either $E = p^2 / 2m$ *or* $p = \sqrt{(2Em)}$ — C1
ratio $= p_e / p_p = \sqrt{(m_e / m_p)}$ — C1
$= \sqrt{\{(9.1 \times 10^{-31}) / (1.67 \times 10^{-27})\}}$
$= 2.3 \times 10^{-2}$ — A1

17.

(a) **(i)** minimum photon energy — B1
minimum energy to remove an electron (from the surface) — B1

(ii) *either* maximum KE is photon energy – work function energy
or max KE when electron ejected from the surface — B1
energies lower than max because energy required to bring electron to the surface — B1

(b) **(i)** threshold frequency = 1.0×10^{15} Hz (*allow* $\pm 0.05 \times 10^{15}$) — C1
work function energy $= hf_0$ — C1
$= 6.63 \times 10^{-34} \times 1.0 \times 10^{15}$
$= 6.63 \times 10^{-19}$ J — A1
(*allow alternative approaches based on use of co-ordinates of points on the line*)

(ii) sketch: straight line with same gradient — M1
displaced to right — A1

(iii) intensity determines number of photons arriving per unit time — B1
intensity determines number of electrons per unit time (not energy) — B1

18.

(a) e.g. no time delay between illumination and emission
max. (kinetic) energy of electron dependent on frequency
max. (kinetic) energy of electron independent of intensity
rate of emission of electrons dependent on/proportional to intensity
(*any three separate statements, one mark each, maximum 3*) — B3

(b) **(i)** (photon) interaction with electron may be below surface — B1
energy required to bring electron to surface — B1

(ii) **1.** threshold frequency = 5.8×10^{14} Hz — A1

2. $\Phi = hf_0$ — C1
$= 6.63 \times 10^{-34} \times 5.8 \times 10^{14}$
$= 3.84 \times 10^{-19}$ (J) — C1
$= (3.84 \times 10^{-19})/(1.6 \times 10^{-19})$
$= 2.4$ eV — A1

or

$hf = \Phi + E_{MAX}$ — (C1)
chooses point on line and substitutes values E_{MAX}, f and h into equation with the units of the hf term converted from J to eV — (C1)
$\Phi = 2.4$ eV — (A1)

19.

(a) photon energy $= hc/\lambda$

$= (6.63 \times 10^{-34} \times 3.0 \times 10^{8})/(590 \times 10^{-9})$ C1

$= 3.37 \times 10^{-19}$ J C1

number $= (3.2 \times 10^{-3})/(3.37 \times 10^{-19})$

$= 9.5 \times 10^{15}$ (allow 9.4×10^{15}) A1

(b) **(i)** $p = h/\lambda$ C1

$= (6.63 \times 10^{-34})/(590 \times 10^{-9})$

$= 1.12 \times 10^{-27}$ kg m s^{-1} C1

total momentum $= 9.5 \times 10^{15} \times 1.12 \times 10^{-27}$

$= 1.06 \times 10^{-11}$ kg m s^{-1} A1

(ii) force $= 1.06 \times 10^{-11}$ N A1

20.

(a) photon 'absorbed' by electron B1

photon has energy equal to difference in energy of two energy levels B1

electron de-excites emitting photon (of same energy) in any direction B1

(b) **(i)** $E = hc/\lambda$ C1

$= (6.63 \times 10^{-34} \times 3 \times 10^{8})/(435 \times 10^{-9})$ C1

$= 4.57 \times 10^{-19}$ J (*allow 2 s.f.*) C1

$= (4.57 \times 10^{-19})/(1.6 \times 10^{-19})$ (eV)

$= 2.86$ eV (*allow 2 s.f.*) A1

(ii) arrow pointing in either direction between –3.41 eV and –0.55 eV B1

21.

(a) discrete amount/packet/quantum of <u>energy</u> M1

of electromagnetic radiation/EM radiation A1

(b) **(i)** $E = hc/\lambda$

$= (6.63 \times 10^{-34} \times 3.0 \times 10^{8})/(570 \times 10^{-9}) = 3.49 \times 10^{-19}$ J A1

(ii) **1.** number $= (2.7 \times 10^{-3})/(3.5 \times 10^{-19})$ C1

$= 7.7 \times 10^{15}$ A1

2. momentum of photon $= h/\lambda$ C1

$= (6.63 \times 10^{-34})/(570 \times 10^{-9})$

$= 1.16 \times 10^{-27}$ kg m s^{-1} C1

change in momentum $= 1.16 \times 10^{-27} \times 7.7 \times 10^{15}$

$= 8.96 \times 10^{-12}$ kg m s^{-1} A1

(*allow E = pc route to* 9×10^{-12})

(c) pressure = (change in momentum per second)/area C1

$= (8.96 \times 10^{-12})/(1.3 \times 10^{-5})$

$= 6.9 \times 10^{-7}$ Pa A1

22.

(a) **(i)** $p = h/\lambda$

$= (6.63 \times 10^{-34})/(6.50 \times 10^{-12})$ C1

$= 1.02 \times 10^{-22}$ N s A1

(ii) $E = hc/\lambda$ *or* $E = pc$

$= (6.63 \times 10^{-34} \times 3.00 \times 10^{8})/(6.50 \times 10^{-12})$ C1

$= 3.06 \times 10^{-14}$ J A1

(b) **(i)** $0.34 \times 10^{-12} = (6.63 \times 10^{-34})/(9.11 \times 10^{-31} \times 3.0 \times 10^{8}) \times (1 - \cos\theta)$ C1

$\theta = 30.7°$ A1

(ii) deflected electron has energy M1

this energy is derived from the incident photon A1

deflected photon has less energy, longer wavelength (so $\Delta\lambda$ always positive) B1

23.

(a) packet/quantum/discrete amount of energy M1

of electromagnetic energy/radiation/waves A1

(b) **(i)** arrow below axis and pointing to right B1

(ii) **1.** $E = hc/\lambda$

$= (6.63 \times 10^{-34} \times 3.0 \times 10^{8})/(6.80 \times 10^{-12})$ C1

$= 2.93 \times 10^{-14}$ J (*accept 2 s.f.*) A1

2. energy of electron $= (3.06 - 2.93) \times 10^{-14}$

$= 1.3 \times 10^{-15}$ J C1

speed $= \sqrt{(2E/m)}$ C1

$= 5.4 \times 10^{7}$ m s^{-1} A1

(c)	momentum is a vector quantity	B1
	either must consider momentum in two directions	
	or direction changes so cannot just consider magnitude	B1

24.

(a)	minimum frequency of e.m. radiation/a photon (not "light")	M1
	for emission of electrons from a surface	A1
	(*reference to light/UV rather than e.m. radiation, allow 1/2*)	
(b)	E_{MAX} corresponds to electron emitted from surface	B1
	electron (below surface) requires energy to bring it to surface, so less than E_{MAX}	B1
(c)	**(i)** $1/\lambda_0 = 1.85 \times 10^6$ *(allow 1.82 to 1.88)*	C1
	$f_0 = c/\lambda_0$	
	$= 3.00 \times 10^8 \times 1.85 \times 10^6$	
	$= 5.55 \times 10^{14}$ Hz	A1
	(ii) $\Phi = hf_0$	
	$= 6.63 \times 10^{-34} \times 5.55 \times 10^{14}$ *(allow ECF from **(c)(i)**)*	C1
	$= 3.68 \times 10^{-19}$ J	A1
(d)	sketch: straight line with same gradient	M1
	intercept between 1.0 and 1.5	A1

25.

(a)	$hc/\lambda = \Phi + E_{MAX}$	M1
	h = Planck constant, c = speed of light/e.m. radiation	A1
(b)	**(i)** gradient of line is hc	M1
	h and c are both constants	A1
	(ii) $\Phi = 2.28 \times 1.6 \times 10^{-19}$	C1
	$= 3.65 \times 10^{-19}$ (J)	
	$hc/\lambda_0 = 3.65 \times 10^{-19}$	
	$\lambda_0 = (6.63 \times 10^{-34} \times 3.0 \times 10^8)/(3.65 \times 10^{-19})$	C1
	$= 5.45 \times 10^{-7}$ m	A1

26.

(a) minimum frequency for electron(s) to be emitted (from surface) M1
reference to frequency of electromagnetic radiation / photon A1

or

frequency causing emission of electron(s) from surface with zero kinetic energy (M1)
reference to frequency of electromagnetic radiation / photon (A1)

(b) (i) positive intercept on ($1/\lambda$)-axis (when extrapolated) B1
straight line with positive gradient B1

(ii) gradient = hc where c is the speed of light B1

(iii) maximum kinetic energy when electron emitted from surface B1
energy is required to bring an electron to the surface B1

(iv) each photon has more energy M1
fewer photons per unit time M1
fewer electrons per unit time / less current A1

27.

(a) electromagnetic radiation/photons incident on a surface B1

causes emission of electrons (from the surface) B1

(b) $E = hc / \lambda$

$= (6.63 \times 10^{-34} \times 3.00 \times 10^{8}) / (436 \times 10^{-9})$ C1

$= 4.56 \times 10^{-19}$ J (4.6×10^{-19} J) A1

(c) (i) $\Phi = hc / \lambda_0$

$\lambda_0 = (6.63 \times 10^{-34} \times 3.00 \times 10^{8}) / (1.4 \times 1.60 \times 10^{-19})$ C1

$= 890$ nm A1

(ii) $\lambda_0 = (6.63 \times 10^{-34} \times 3.00 \times 10^{8}) / (4.5 \times 1.60 \times 10^{-19})$

$= 280$ nm A1

(d) caesium:
wavelength of photon less than threshold wavelength (or v.v.)
or
λ_0 = 890 nm > 436 nm
so yes — A1

tungsten:
wavelength of photon greater than threshold wavelength (or v.v.)
or
λ_0 = 280 nm < 436 nm
so no — A1

28.

(a) (i) electron diffraction/electron microscope (allow other sensible suggestions) — B1

(ii) photoelectric effect/Compton scattering (allow other sensible suggestions) — B1

(b) (i) arrow clear from –0.54 eV to –3.40 eV — B1

(ii) $E = hc/\lambda$ *or* $E = hf$ and $c = f\lambda$ — C1

$\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^{8}) / [(3.40 - 0.54) \times 1.60 \times 10^{-19}] = 4.35 \times 10^{-7}$ m — A1

(c) (i) wavelength associated with a particle — M1
that is moving/has momentum/has speed/has velocity — A1

(ii) $\lambda = h/mv$

$v = (6.63 \times 10^{-34}) / (9.11 \times 10^{-31} \times 4.35 \times 10^{-7})$ — C1

$= 1.67 \times 10^{3}\,\mathrm{m\,s^{-1}}$ — A1

29.

10(a)	packet / quantum of energy	**M1**
	of electromagnetic radiation	**A1**
10(b)(i)	light is re-emitted in all directions / only part of the re-emitted light is in the direction of the beam	**B1**
10(b)(ii)	an arrow between –3.40 eV and –1.51 eV and an arrow between –3.40 eV and –0.85 eV	**B1**
	all arrows shown point 'upwards'	**B1**
10(b)(iii)	$E = hc/\lambda$ *or* $E = hf$ and $c = f\lambda$	**C1**
	$2.60 \times 1.60 \times 10^{-19} = (6.63 \times 10^{-34} \times 3.00 \times 10^{8}) / \lambda$	**C1**
	$\lambda = 4.8 \times 10^{-7}$ m	**A1**

30.

11(a)	electrons (in gas atoms/molecules) interact with photons	**B1**
	photon energy causes electron to move to higher energy level/to be excited	**B1**
	photon energy = difference in energy of (electron) energy levels	**B1**
	when electrons de-excite, photons emitted in all directions (so dark line)	**B1**
11(b)(i)	photon energy $\propto 1 / \lambda$	**C1**
	energy = 1.68 eV	**A1**
	or	
	$E = hc / \lambda$ $E = 6.63 \times 10^{-34} \times 3.0 \times 10^{8} / (740 \times 10^{-9})$ $= 2.688 \times 10^{-19}$ J	**(C1)**
	energy = 1.68 eV	**(A1)**
11(b)(ii)	3.4 eV → 1.5 eV 3.4 eV → 0.85 eV 3.4 eV → 0.54 eV *all correct and none incorrect 2/2* *2 correct and 1 incorrect or only 2 correctly drawn 1/2*	**B2**

31.

10(a)	two from: • frequency below which electrons not ejected • <u>maximum</u> energy of electron depends on frequency • <u>maximum</u> energy of electrons does not depend on intensity • instantaneous emission of electrons	**B2**
10(b)(i)	(λ_0 is the) threshold wavelength **or** wavelength corresponding to threshold frequency **or** maximum wavelength for emission of electrons	**B1**
10(b)(ii)1.	intercept = $1/\lambda_0 = 2.2 \times 10^{6}\ \text{m}^{-1}$ $\lambda_0 = 4.5 \times 10^{-7}$ m or 450 nm	**A1**
10(b)(ii)2.	gradient = hc	**C1**
	gradient = 2.0×10^{-25} or correct substitution into gradient formula	**C1**
	$h = (2.0 \times 10^{-25}) / (3.0 \times 10^{8}) = 6.7 \times 10^{-34}$ J s	**A1**
10(c)	line: same gradient	**B1**
	straight line, positive gradient, intercept at greater than 2.2×10^{6} when candidate's line extrapolated	**B1**

32.

11(a)	packet/quantum of energy of electromagnetic/EM radiation	**B1**
11(b)(i)	$E = hf$ $1.1 \times 10^6 \times 1.60 \times 10^{-19} = 6.63 \times 10^{-34} \times f$	**C1**
	$f = 2.7 \times 10^{20}$ (2.65×10^{20}) Hz	**A1**
11(b)(ii)	$p = h/\lambda = hf/c$ $= (6.63 \times 10^{-34} \times 2.65 \times 10^{20})/(3.00 \times 10^8)$ **or** $p = E/c$ $= (1.1 \times 1.60 \times 10^{-13})/(3.00 \times 10^8)$	**C1**
	$p = 5.9 \times 10^{-22}$ (5.87×10^{-22}) N s	**A1**
11(c)	$123 \times 1.66 \times 10^{-27} \times v = 5.87 \times 10^{-22}$	**C1**
	$v = 2.9 \times 10^3$ m s^{-1}	**A1**

33.

10(a)	λ_0 marked and graph line passing through $E_{MAX} = 0$ at $\lambda = \lambda_0$	**B1**
	graph line with λ always $< \lambda_0$	**B1**
	negative gradient with correct concave curvature	**B1**
10(b)	curve with negative gradient and correct concave curvature	**M1**
	not touching either axis	**A1**

34.

10(a)	emission of electron	**B1**
	when electromagnetic radiation incident (on surface)	**B1**
10(b)(i)	packet/quantum/discrete amount of <u>energy</u>	**M1**
	of electromagnetic radiation	**A1**
10(b)(ii)	$E = hc/\lambda$	**C1**
	$= (6.63 \times 10^{-34} \times 3.00 \times 10^8)/(420 \times 10^{-9})$ $= 4.7 \times 10^{-19}$ J	**A1**
10(b)(iii)	sodium: yes zinc: no	**B1**

35.

11(a)	discrete amount/quantum/packet of <u>energy</u>	**M1**
	of electromagnetic radiation	**A1**
11(b)(i)	energy $= hc/\lambda$	**C1**
	$\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^8)/(0.51 \times 10^6 \times 1.60 \times 10^{-19})$ $= 2.4 \times 10^{-12}$ m	**A1**

11(b)(ii)	$p = h/\lambda$ $= (6.63 \times 10^{-34})/(2.44 \times 10^{-12})$ **or** $p = E/c$ $= (0.51 \times 1.60 \times 10^{-13})/(3.00 \times 10^{8})$	**C1**
	$p = 2.7 \times 10^{-22}$ N s	**A1**
11(c)(i)	$E = c^2\Delta m$	**C1**
	$\Delta m = (0.51 \times 1.60 \times 10^{-13})/(3.00 \times 10^{8})^2$ $= 9.1 \times 10^{-31}$ kg	**A1**
11(c)(ii)	(momentum is conserved so) nucleus must have momentum in opposite direction to photon	**B1**

36.

11(a)	discrete amount/quantum/packet of <u>energy</u>	**M1**
	of electromagnetic radiation	**A1**
11(b)	mostly dark/dark background	**B1**
	coloured lines	**B1**
11(c)(i)	6	**A1**
11(c)(ii)	**1.** maximum photon energy = 13.6 – 0.85 (= 12.75 eV)	**C1**
	maximum kinetic energy = (13.6 – 0.85) – 5.6 = 7.2 eV	**A1**
	2. energy = hc/λ	**C1**
	$\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^{8}) / [(13.6 - 0.85) \times 1.60 \times 10^{-19}]$	**C1**
	$= 9.8 \times 10^{-8}$ m	**A1**

37.

11(a)	Any three points from: • (max) energy of emitted electrons depends on frequency • (max) energy of emitted electrons does not depend on intensity • rate of emission of electrons depends on intensity (at constant frequency) • existence of frequency below which no emission of electrons • instantaneous emission of electrons • increasing the frequency at constant intensity decreases the rate of emission of electrons	**B3**
11(b)(i)	photon energy = hc/λ	**C1**
	$= (6.63 \times 10^{-34} \times 3.0 \times 10^{8}) / (380 \times 10^{-9})$ $(= 5.23 \times 10^{-19}$ J)	**C1**
	= 3.3 eV	**A1**
11(b)(ii)	photon energy must be greater than work function (energy)	**B1**
	so sodium and calcium	**B1**
11(c)	$\lambda = h/p$	**C1**
	$p = (6.63 \times 10^{-34}) / (380 \times 10^{-9})$ $= 1.74 \times 10^{-27}$ N s	**C1**
	force $= 1.74 \times 10^{-27} \times 7.6 \times 10^{14}$ $= 1.3 \times 10^{-12}$ N	**A1**

38.

11(a)	packet/quantum of <u>energy</u>	**M1**
	of electromagnetic radiation	**A1**
11(b)(i)	$E = hc / \lambda$	**C1**
	$1.18 \times 1.60 \times 10^{-13} = (6.63 \times 10^{-34} \times 3.00 \times 10^{8}) / \lambda$ $\lambda = 1.05 \times 10^{-12}$ m	**A1**
11(b)(ii)	$\lambda = h / p$ **or** $E = pc$	**C1**
	$p = (6.63 \times 10^{-34}) / (1.05 \times 10^{-12})$ **or** $p = (1.18 \times 1.60 \times 10^{-13}) / (3.00 \times 10^{8})$ leading to $p = 6.3 \times 10^{-22}$ N s	**B1**
11(c)	$6.3 \times 10^{-22} = 60 \times 1.66 \times 10^{-27} \times v$	**C1**
	$v = 6.3 \times 10^{3}$ m s^{-1}	**A1**

39.

11(a)	energy (of photon) required to remove electron	**M1**
	from a surface **or** reference to <u>minimum</u> energy **or** reference to zero <u>kinetic</u> energy	**A1**
11(b)(i)	**1.** photon energy $= hc / \lambda$	**C1**
	$= (6.63 \times 10^{-34} \times 3.00 \times 10^{8}) / (280 \times 10^{-9})$ $= 7.1 \times 10^{-19}$ J	**A1**
	2. electron energy $= (7.1 - 5.5) \times 10^{-19}$ J	**C1**
	$\frac{1}{2} \times 9.11 \times 10^{-31} \times v^2 = (7.1 - 5.5) \times 10^{-19}$	**C1**
	$v = 5.9 \times 10^{5}$ m s^{-1}	**A1**
11(b)(ii)	energy is required to bring electron to the surface	**B1**
11(c)	no change / decreases increases / decreases	**B4**

40.

10(a)	energy of a photon required to remove an electron	**B1**
	either: energy to remove electron from a surface *or*: <u>minimum</u> energy to remove electron *or*: energy to remove electron with zero <u>kinetic</u> energy	**B1**
10(b)(i)	Correct read off from graph of f as 5.45×10^{14} Hz when $E_{MAX} = 0$ $5.45 \times 10^{14} \times 6.63 \times 10^{-34}$	**C1**
	$= \mathbf{3.6 \times 10^{-19}}$ J	**A1**
10(b)(ii)	$3.6 \times 10^{-19} / 1.6 \times 10^{-19} = 2.3$ eV so potassium	**A1**
10(c)(i)	each photon has same energy so no change	**B1**
10(c)(ii)	more photons (per unit time) so (rate of emission) increases	**B1**

41.

11(a)(i)	$E = mc^2$	**C1**
	$= 9.11 \times 10^{-31} \times (3.0 \times 10^8)^2$ $= 8.2 \times 10^{-14}$ J	**A1**
11(a)(ii)	$p = h / \lambda$ **and** $E = hc / \lambda$ **or** $E = pc$	**C1**
	$p = (8.2 \times 10^{-14}) / (3.0 \times 10^8)$ $= 2.7 \times 10^{-22}$ N s	**A1**
11(b)	total momentum (before and after interaction) is zero **or** momentum must be conserved (in the interaction) **or** momentum of the photons must be equal and opposite	**B1**
	(photons emitted in) opposite directions	**B1**

42.

11(a)(i)	quantum of energy	**M1**
	of electromagnetic radiation	**A1**
11(a)(ii)	arrow (on Fig. 11.1) pointing upwards and to the right	**B1**
11(b)(i)	$\lambda = h / p$	**C1**
	$p = (6.63 \times 10^{-34}) / (544 \times 10^{-9})$ $= 1.22 \times 10^{-27}$ N s	**A1**
11(b)(ii)	energy $= hc / \lambda$	**C1**
	$= 6.63 \times 10^{-34} \times 3.00 \times 10^8 \times (540^{-1} - 544^{-1}) \times 10^9$ $= 2.7 \times 10^{-21}$ J	**A1**
11(c)	(smaller wavelength corresponds to) greater photon energy	**B1**
	any one point from: • (deflected) photon loses energy (so not possible) • (deflected) photon would need to gain energy (so not possible) • electron would need to lose energy (so not possible) • initially electron energy is zero (so not possible)	**B1**

43.

11(a)	any two points from: • (maximum) kinetic energy of electrons is independent of intensity • maximum kinetic energy of electrons depends on frequency • no time delay (between illumination and emission)	**B2**
11(b)(i)	(for $E_{MAX} = 0$,) $1 / \lambda_0 = 1.93 \times 10^6$ (m^{-1})	**C1**
	$f_0 = 3.00 \times 10^8 \times 1.93 \times 10^6$ $= 5.8 \times 10^{14}$ Hz	**A1**
11(b)(ii)	$hc / \lambda = \Phi + E_{MAX}$	**C1**
	hc = gradient	**C1**
	gradient = e.g. $[(0.40 - 0.20) \times 1.60 \times 10^{-19}] / [(2.25 - 2.09) \times 10^6]$ (*working needed*) ($= 2.0 \times 10^{-25}$)	**M1**
	$h = (2.0 \times 10^{-25}) / (3.00 \times 10^8) = 6.7 \times 10^{-34}$ J s (*both working and answer needed*)	**A1**
11(c)	straight line with same gradient as the original	**B1**
	straight line with x-axis intercept greater than $1.93 \times 10^6\ \text{m}^{-1}$	**B1**

44.

12(a)	• frequency determines energy of photon • intensity determines number of photons (per unit time) • intensity does not determine energy of a photon *Any two points, 1 mark each*	**B2**
	kinetic energy (of the electron) depends on the energy of one photon	**B1**
12(b)(i)	$E = hc / \lambda$ **or** $E = hf$ **and** $c = f\lambda$	**C1**
	$E = (6.63 \times 10^{-34} \times 3.00 \times 10^{8}) / (250 \times 10^{-9})$	**C1**
	$(= 7.96 \times 10^{-19}\ \text{J})$ $= 5.0\ \text{eV}$	**A1**
12(b)(ii)	E_{MAX} = photon energy – work function	**C1**
	work function = 5.0 – 1.4 = 3.6 eV	**A1**

45.

12(a)	quantum of energy	**M1**
	of electromagnetic radiation	**A1**
12(b)(i)	energy = hc / λ **or** energy = hf **and** $f = c / \lambda$	**C1**
	$0.57 \times 10^{6} \times 1.60 \times 10^{-19} = (6.63 \times 10^{-34} \times 3.00 \times 10^{8}) / \lambda$ $\lambda = 2.2 \times 10^{-12}\ \text{m}$	**A1**
12(b)(ii)	$p = h / \lambda$	**C1**
	$= (6.63 \times 10^{-34}) / (2.2 \times 10^{-12})$ $= 3.0 \times 10^{-22}\ \text{N s}$	**A1**
	or	
	$p = E / c$	**(C1)**
	$= (0.57 \times 10^{6} \times 1.60 \times 10^{-19}) / (3.00 \times 10^{8})$ $= 3.0 \times 10^{-22}\ \text{N s}$	**(A1)**
12(c)(i)	mass (of Sm-157 nucleus) = $157 \times 1.66 \times 10^{-27}$ **or** mass (of Sm-157 nucleus) = $0.157 / (6.02 \times 10^{23})$	**C1**
	recoil speed = $(3.00 \times 10^{-22}) / (157 \times 1.66 \times 10^{-27})$ $= 1.2 \times 10^{3}\ \text{m s}^{-1}$	**A1**
12(c)(ii)	$(1.2 \times)\ 10^{3}\ \text{m s}^{-1}$ is <u>much</u> less than $(3.0 \times)\ 10^{8}\ \text{m s}^{-1}$	**B1**

46.

10(a)(i)	photoelectric effect	**B1**
10(a)(ii)	electron diffraction	**B1**
10(b)(i)	$\lambda = h / p$	**M1**
	h is the Planck constant	**A1**
10(b)(ii)	de Broglie (wavelength)	**B1**
10(c)(i)	$½mv^2 = eV$	**C1**
	$½ \times 9.11 \times 10^{-31} \times v^2 = 1.60 \times 10^{-19} \times 4800$ <u>so</u> $v = 4.1 \times 10^7$ m s^{-1}	**A1**
10(c)(ii)	$\lambda = h / mv$ $= 6.63 \times 10^{-34} / (9.11 \times 10^{-31} \times 4.1 \times 10^7)$	**C1**
	$= 1.8 \times 10^{-11}$ m	**A1**

47.

9(a)(i)	emission of electrons (from a metal surface)	**B1**
	when electromagnetic radiation is incident (on electrons)	**B1**
9a(ii)	<u>minimum</u> energy required for an electron to leave surface	**B1**
9(b)(i)	threshold (frequency)	**B1**
9(b)(ii)	• photons are (discrete) packets of energy • energy of photons depends on frequency (of EM radiation) • electrons can only absorb a single photon (of energy) *Any two points, 1 mark each*	**B2**
	emission only possible if photon energy is at least the work function	**B1**
9(b)(iii)	work function = hf_0 $= 6.63 \times 10^{-34} \times 6.93 \times 10^{14}$	**C1**
	$= 4.59 \times 10^{-19}$ (J) $= 4.59 \times 10^{-19} / 1.60 \times 10^{-19}$ (eV) $= 2.87$ eV	**A1**

48.

8(a)	$\lambda = \frac{h}{p}$ or $\lambda = \frac{h}{mv}$	**M1**
	where h is the Planck constant **and** p is the momentum (of particle) / mv is the momentum (of particle) / m is the mass (of particle) and v is the velocity (of particle)	**A1**
8(b)(i)	(electron) diffraction	**B1**
8(b)(ii)	moving electrons behave like waves	**B1**
8(b)(iii)	spacing between atoms ≈ wavelength of electron *or* diameter of atom ≈ wavelength of electron	**B1**
8(b)(iv)	Any one of: • wavelength has decreased • electron had greater momentum	**M1**
	so (accelerating) p.d. was increased	**A1**

49.

7(a)	quantum of energy	**M1**
	of electromagnetic radiation	**A1**
7(b)(i)	photoelectric effect	**B1**
7(b)(ii)	• there is a frequency below which no electrons are emitted **or** threshold frequency = 5.4×10^{14} Hz • work function of the metal = 3.6×10^{-19} J (or 2.2 eV) • E_{MAX} increases (linearly) with (increasing) frequency • gradient of the line is the Planck constant **or** gradient of the line is 6.7×10^{-34} J s *Any three bullet points, 1 mark each*	**B3**
7(c)(i)	different threshold frequency	**B1**
	(line has) same gradient but different intercept	**B1**
7(c)(ii)	photons have same energy	**B1**
	line unchanged	**B1**

50.

8(a)(i)	photoelectric effect	**B1**
8(a)(ii)	electron diffraction	**B1**
8(b)(i)	$\lambda = h / p$	**C1**
	$p = 4 \times 1.66 \times 10^{-27} \times 6.2 \times 10^{7}$ $(= 4.1 \times 10^{-19}$ N s)	**C1**
	$\lambda = 6.63 \times 10^{-34} / 4.1 \times 10^{-19}$ $= 1.6 \times 10^{-15}$ m	**A1**
8(b)(ii)	line with negative gradient throughout	**B1**
	curve asymptotic to both axes with non-zero λ at $v = 6.2 \times 10^{7}$ m s^{-1}	**B1**
8(c)	(de Broglie) wavelength negligible compared with width of doorway	**B1**

51.

8(a)	photon energy (to remove electron)	**B1**
	minimum energy to remove electron **or** energy to remove electron from surface **or** energy to remove electron with zero kinetic energy	**B1**
8(b)(i)	photon energy = hf	**C1**
	number per unit time = $8.36 \times 10^{-3} / (1.36 \times 10^{15} \times 6.63 \times 10^{-34})$ $= 9.27 \times 10^{15}$ s^{-1}	**A1**
8(b)(ii)	$hf = \Phi + E_{MAX}$	**C1**
	$\Phi = (1.36 \times 10^{15} \times 6.63 \times 10^{-34}) - (3.09 \times 10^{-19})$ $= 5.93 \times 10^{-19}$ J	**A1**
8(c)(i)	greater photon energy (and same work function)	**M1**
	so maximum kinetic energy is increased	**A1**

8(c)(ii)	(greater photon energy and same power so) lower number of photons (per unit time)	**M1**
	(each electron absorbs one photon) so lower rate of emission	**A1**

52.

9(a)(i)	• energy of photon has a corresponding frequency • change in electron energy level emits a single photon • photon energy = difference in energy levels • discrete frequencies must have come from discrete energy gaps • discrete energy changes imply discrete energy levels *Any three points, 1 mark each*	**B3**
9(a)(ii)	transition (to – 3.400 eV) from X corresponds to 658 nm line	**C1**
	$E_1 - E_2 = hc / \lambda$	**C1**
	$E_1 - (-3.400) = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (658 \times 10^{-9} \times 1.60 \times 10^{-19})$ and so $E_1 = -1.51$ eV (*full substitution and answer needed*)	**A1**
9(b)(i)	redshift	**B1**
9(b)(ii)	moving away (from observer)	**B1**
9(b)(iii)	$\Delta\lambda / \lambda = v / c$ e.g. for 658 nm line: $\Delta\lambda = 686 - 658$ (= 28 nm) (*other lines may be used*)	**C1**
	$28 / 658 = v / (3.00 \times 10^8)$ (*other lines may be used*)	**C1**
	$v = 1.3 \times 10^7$ m s^{-1}	**A1**
9(c)	$v = H_0 d$	**C1**
	$H_0 = (1.3 \times 10^7) / (5.7 \times 10^{24})$ $= 2.3 \times 10^{-18}$ s^{-1}	**A1**

53.

7(a)	photon absorbed (by electron) and electron excited	**B1**
	photon energy equal to difference in (energy of two) energy levels	**B1**
	photon energy relates to a single wavelength / single frequency	**B1**
	electron de-excites and emits photon in any direction	**B1**
7(b)	$\frac{hc}{\lambda} = \Delta E$	**C1**
	uses 658nm	**C1**
	$\frac{6.63\times10^{-34}\times3.00\times10^{8}}{658\times10^{-9}} = -E_1 - (-3.40 \times 1.60 \times 10^{-19})$ $E_1 = -2.42 \times 10^{-19}$ J	**A1**

54.

7(a)	wavelength associated with a <u>moving</u> particle	**B1**
7(b)(i)	(electron) diffraction	**B1**
7(b)(ii)	beam spreads out indicating diffraction **or** light and dark regions indicate an interference pattern	**B1**
	electron beam is behaving as a wave	**B1**
7(c)(i)	central blob and concentric rings	**B1**
	rings closer together (than previously)	**B1**
7(c)(ii)	(greater p.d. so) electrons to have greater momentum	**B1**
	greater momentum so decrease in (de Broglie) wavelength	**B1**
	lower (de Broglie) wavelength (for same grating spacing in crystal) causes: smaller diffraction angle **or** smaller angle of intensity maxima (for each order) **or** decrease in fringe spacing in diffraction pattern	**B1**

55.

8(a)	transition (emits) (one) photon with energy equal to the difference in energy between the two levels	**B1**
	frequency of radiation corresponds to energy of photon	**B1**
8(b)(i)	line to the left of the pair in Fig. 8.2, labelled A	**B1**
	larger gap between line A and the nearest of the pair in Fig. 8.2 than between the lines in the pair	**B1**
8(b)(ii)	line to the left of both the pair in Fig. 8.2 and line A, labelled B	**B1**
	larger gap between line B and line A than between line A and the nearest one of the pair in Fig. 8.2	**B1**
8(c)	$E = hf$	**C1**
	$E_3 = E_1 + h(f_A + f_B)$	**A1**

56.

8(a)(i)	$p = E / c$	**M1**
	$E = hc / \lambda$ and completion of algebra leading to $p = h / \lambda$	**A1**
8(a)(ii)	wavelength = $(6.63 \times 10^{-34}) / (9.5 \times 10^{-28}) = 700 \times 10^{-9}$ m so red	**B1**
8(b)(i)	power = intensity × area	**C1**
	number per unit time = $(160 \times 2.5 \times 10^{-6}) / (9.5 \times 10^{-28} \times 3.00 \times 10^{8}) = 1.4 \times 10^{15}\ \text{s}^{-1}$	**A1**
8(b)(ii)	pressure = force / area	**C1**
	force = rate of change of momentum $= 2 \times 9.5 \times 10^{-28} \times 1.4 \times 10^{15}$	**C1**
	pressure = $(2 \times 9.5 \times 10^{-28} \times 1.4 \times 10^{15}) / (2.5 \times 10^{-6})$ $= 1.1 \times 10^{-6}$ Pa	**A1**
8(c)	photons have greater momentum **or** fewer photons per unit time	**B1**
	greater photon momentum but smaller number of photons (per unit time) so pressure is the same	**B1**

57.

8(a)	packet / quantum of <u>energy</u>	**M1**
	of electromagnetic radiation	**A1**
8(b)(i)	photoelectric effect	**B1**
8(b)(ii)	• electron needs a minimum energy to escape **or** electron emitted if energy in packet is enough • energy must be absorbed in packets that are related to frequency • intensity relates to number of packets (not to energy in packet) • electron absorbs only a single whole packet *Any three points, 1 mark each*	**B3**
8(c)(i)	Planck constant	**B1**
8(c)(ii)	– work function (energy)	**B1**